

Which Pair Is It About?

Triple-Conditioned Verification of Literature-Mined Biotic Interactions

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Abstract

Literature-mining pipelines that populate biotic-interaction databases propose candidate (taxon, relation, taxon) triples by co-occurrence, then decide which to keep with a *sentence-level* classifier: does this passage describe an interaction? That is the wrong question. A candidate is retrieved precisely because its passage already names two taxa and an interaction term, so what needs deciding is whether the passage asserts *this* relation between *these two*. We test the claim with a controlled comparison: one encoder, one 48k-row teacher-labelled corpus, three seeds, differing only in whether the model sees the candidate triple. On 437 expert-graded candidates the triple-conditioned verifier raises per-seed AUPRC from 0.851 to 0.910 and F1 from 0.775 to 0.860 (76 items fixed against 30 broken, McNemar $p = 1.2 \times 10^{-5}$), dominating the sentence-level model at every threshold. The gain is where the argument puts it: +0.095 AUPRC on passages that name three or more taxa, and none at all on passages that name two. We decompose the residual false positives by mechanism, show that eight deterministic candidate rules validated on training data before evaluation raise precision from 0.825 to 0.878, and report the deployed two-head configuration: F1 0.909 at a pre-specified threshold, at 33 candidates per second on a CPU.

1 Introduction

Biotic-interaction databases such as GloBI (Poe-[len et al., 2014](#)) are populated in part by text mining. A retrieval system proposes candidate triples (s_1, r, s_2) together with the passage that allegedly supports them, and something must decide which candidates are worth a curator’s attention. The natural choice, and the common one, is a sentence-level interaction classifier: show it the passage and ask whether it describes a biotic interaction.

That question is answered before it is asked. Candidates are proposed by co-occurrence: a passage

naming two taxa near an interaction term yields a candidate whether or not it asserts that interaction between those two. Every candidate therefore arrives already satisfying, lexically, the property a sentence-level classifier was trained to detect — in the pipeline we study, on all 175,588 candidates of its retrieval pool (§2). What actually needs deciding is narrower: whether the passage asserts *this* relation between *these two* taxa. When a passage names three organisms, it can support one candidate and refute the other two, and a classifier that sees only the passage must give all three the same answer.

We make this argument testable rather than rhetorical. We train the same 110M-parameter encoder on the same teacher-labelled corpus twice, once reading the passage alone and once reading the candidate triple as a query, and compare them on a 437-item benchmark graded by a domain expert (§4). Everything except the input is held fixed, so the difference between the two is attributable to the formulation and to nothing else. The triple-conditioned model wins by a wide and significant margin, and the gain is located exactly where the argument predicts: it is +0.095 AUPRC on passages naming three or more taxa and −0.002 on passages naming two or fewer.

Contributions.

- A controlled comparison of the sentence-level and triple-conditioned formulations on one encoder and one corpus, with the gain located where the argument says it should be (§4).
- A mechanism analysis of the errors that survive, including a bound showing that checking the triple’s components separately cannot reach the largest class (§5).
- Deterministic candidate rules, each validated on 42k training rows before evaluation, that raise precision by 0.053 at a recall cost of 0.020 (§6).

- A deployed two-head configuration, its threshold fixed in advance, running on a CPU (§7), and four negative results (Appendix B).

2 Setting

Where candidates come from. BiotXplorer (Ruch et al., 2024) proposes candidate biotic-interaction triples from the literature indexed by SIB Literature Services (Gobeill et al., 2020). For each passage it matches taxon surface forms against a taxonomy and interaction surface forms against a vocabulary derived from the Relation Ontology, and emits every resulting (s_1, r, s_2) combination with the passage, the matched surface strings, and their canonical forms. This is the application we build for, and we use it only as the source of candidates: no component of it is a baseline in this paper.

Why the sentence-level question is constant here.

We pulled the retrieval pool in full: 175,588 candidate rows over 165,325 distinct passages and 115,404 distinct triplet keys. Both taxon surface forms and the interaction surface form are non-empty on all 175,588 rows, because the presence of an interaction term is the pool’s construction filter, and an independent lexical scan for GloBI interaction vocabulary fires on 99.82% of the same passages. A classifier trained to separate passages with interaction language from passages without it is, on this distribution, being asked about a constant. The missing input is not an annotator artefact but the query the pipeline itself issued.

Benchmark. One benchmark of 437 rows, 246 positive, in three blocks of different provenance; we report every block rather than pooling them into a number that hides their disagreement. BIOTX100 is retrieved candidates, each the top-ranked candidate for its passage, graded by a domain expert on four axes: whether each taxon surface form was correctly resolved, whether the relation was correct, and whether the whole triple is supported by its passage. After removing three training duplicates it contributes 97 rows, 75 positive. REJECT50 is candidates the retrieval pipeline discarded, curated for the ones it should have kept: 41 clean rows, 9 positive. It is the paper’s recall-side instrument. TEST299 is 299 pair-level items from expert curation, 0.542 positive, and is where passages most often name several taxa.

The blocks do not carry the same gold.

BIOTX100 is the only block graded at the level of the whole triple; TEST299 was curated pair-level and its stored relation term was attached by retrieval rather than by an annotator. We therefore evaluate everywhere at the level all three share — *do these two taxa interact in this passage* — and use the triple-level grades only to diagnose (§5). This understates what triple conditioning buys, since a candidate with the right pair and the wrong relation counts as correct, and we prefer understating it to scoring two blocks against a relation term nobody checked.

Query representation. Each candidate carries a canonical string and the surface form matched in the passage. On BIOTX100 these differ for s_1 , s_2 and r on 34, 38 and 59 of 100 rows, and the canonical string is absent from the passage on 31, 27 and 54. We always query with the surface form, which is what the model can see.

Evaluation protocol. Thresholds are where this kind of evaluation goes wrong most easily, so the policy is fixed in advance. A model’s reported operating point is either **pre-specified** — chosen on its own training-development split, never on the benchmark — or **block-held-out**: fitted on two of the three source blocks and applied to the third, so that every row is scored by a threshold that never saw it and every row is used. Comparisons between models use the same policy for both. Intervals are 20,000-sample bootstraps or Clopper–Pearson; McNemar tests carry a continuity correction. Where three seeds exist we report the per-seed mean and standard deviation, and where we report the average of three checkpoints’ probabilities we call it an ensemble. The benchmark ships an `in_train` flag on 9 rows; a 5-gram Jaccard scan against all 48,338 training passages finds 3 more at Jaccard 1.000, all in BIOTX100. All 12 are excluded throughout.

3 Two formulations, one corpus

The teacher. Labels come from a locally hosted 32B open-weight instruction model (Qwen Team, 2025), prompted zero-shot at temperature 0 and asked, per candidate, whether the passage supports *this* triple. Its labels therefore vary across candidates that share a passage. Against 150 human-graded items its mislabel rate is $22/150 = 14.7\%$, Clopper–Pearson [9.4%, 21.4%], split 10 false positives and 12 false negatives.

System	AUPRC	P	R	F1	Query given to the verifier	P	R	F1	AUPRC
Accept everything	0.563	0.563	1.000	0.720	Retrieved triple	0.842	0.890	0.866	0.912
Sentence-level baseline	0.851 $\pm$.006	0.730	0.825	0.775	Generic relation, pair kept	0.876	0.687	0.770	0.874
Triple-conditioned verifier	0.910$\pm$.006	0.825	0.898	0.860	Relation shuffled across items	0.879	0.679	0.766	0.897
					Relation only, no taxa	0.917	0.045	0.085	0.677
					Empty query	0.653	0.191	0.296	0.641

Table 1: The controlled comparison on 437 rows: one encoder, one corpus, only the input differs. AUPRC is the per-seed mean $\pm$ standard deviation over three seeds; P/R/F1 are three-checkpoint ensembles at block-held-out thresholds. The verifier fixes 76 items the baseline gets wrong and breaks 30 it gets right (McNemar $p = 1.2 \times 10^{-5}$).

Passages	n	Sentence	Triple	Gain
naming $\leq$ 2 taxa	161	0.876	0.874	-0.002
naming $\geq$ 3 taxa	276	0.846	0.941	+0.095
TEST299	299	0.819	0.925	+0.106
BIOTX100	97	0.954	0.939	-0.015
REJECT50	41	0.465	0.497	+0.032

Table 2: Where the gain lives, AUPRC of the two ensembles. Taxa are counted as distinct mentions found by a biodiversity NER model (en_ner_eco_biobert) in the passage. TEST299 passages name 5.1 taxa on average, BIOTX100 2.6 and REJECT50 1.7.

The corpus. 48,338 teacher-labelled rows, 0.343 positive, drawn from the same retrieval pool as the benchmark (disjoint passages): a 20,000-row base sample, 16,546 *pair-contrast* rows, 6,201 actively sampled rows, and 5,591 rows with the candidate’s argument order reversed. The pair-contrast block is what gives the corpus its point: 2,017 passages enter as several rows that are identical in the passage and differ only in the candidate and the label.

The two models. Both are BiomedBERT-base cross-encoders (110M parameters), trained for three epochs on that corpus with the same optimiser and schedule, three seeds each, with a development split grouped on taxon pair so that no pair spans training and development. They differ in one respect. The **sentence-level baseline** reads the passage alone. It is the formulation a pipeline reaches for first, and on rows that share a passage it is shown the same input with different labels, which it cannot fit. The **triple-conditioned verifier** reads the candidate s_1 [SEP] r [SEP] s_2 as segment A and the passage as segment B (Nogueira and Cho, 2019), so its input moves when the candidate moves.

Table 3: Degrading segment A of the trained verifier at inference, weights and threshold ($\tau = 0.04$) unchanged. Base rate 0.563. Per-seed standard deviations on AUPRC are ≤ 0.027 .

4 Results

4.1 Conditioning on the triple

Table 1 gives the comparison. Conditioning on the candidate triple raises per-seed AUPRC from 0.851 $\pm$ 0.006 to 0.910 $\pm$ 0.006 and, at block-held-out thresholds, F1 from 0.775 to 0.860 — precision and recall both. On decision correctness the verifier fixes 76 items and breaks 30, McNemar $p = 1.2 \times 10^{-5}$. The advantage is not an artefact of where we chose to cut: the verifier dominates the baseline at every threshold (Figure 1), and at $\tau = 0.25$, for example, holds precision 0.899 at recall 0.793 where the baseline holds 0.837 at 0.687.

Table 2 locates the gain, and it is the most direct test of the paper’s argument. Split by how many distinct taxa a passage names, the two formulations are indistinguishable when it names two or fewer — AUPRC 0.876 against 0.874 — because then there is only one pair the passage could be about, and asking whether it describes an interaction is asking the right question. When it names three or more, conditioning on the triple is worth +0.095. The blocks follow the same pattern for the same reason: TEST299, whose passages name 5.1 taxa on average, gains 0.106, while BIOTX100, at 2.6 and every row the top-ranked candidate of its passage, gains nothing.

4.2 The query, not the encoder

A model that reads the triple could in principle be winning on something other than the triple. Degrading the query at inference, with weights and threshold unchanged, shows it is not (Table 3). Deleting both taxa collapses ranking from 0.912 to 0.677 against a base rate of 0.563, and emptying the query entirely gives 0.641. Replacing the retrieved relation with a generic one, pair kept, costs 0.096 F1; shuffling the relation across items costs 0.100. The decision is carried overwhelmingly by the pair and materially by the relation.

τ	keep	triple-conditioned			sentence-level		
		P	R	F1	P	R	F1
0.01	290	0.779	0.919	0.843	0.690	0.915	0.787
0.04	260	0.842	0.890	0.866	0.763	0.813	0.787
0.10	238	0.866	0.837	0.851	0.796	0.732	0.763
0.25	217	0.899	0.793	0.842	0.837	0.687	0.754
0.50	181	0.923	0.679	0.782	0.866	0.524	0.653
0.90	133	0.947	0.512	0.665	0.930	0.163	0.277

Table 4: Both models’ operating curves on all 437 rows, three-checkpoint ensembles. Rows are shown to make the shape visible; the reported operating points are those of Table 1.

The empty-query arm is *not* the sentence-level baseline: it is a model trained with the query and deprived of it, and it does far worse (0.641) than a model trained without the query from the start (0.851). That gap is a warning about a common shortcut. Ablating an input at inference measures dependence on it; it does not stand in for training without it.

Neither does the gain come from the corpus’s pair-contrast design. A verifier trained on the 20,000-row base sample alone, with the same triple-conditioned input, is indistinguishable from the full one: per-seed AUPRC 0.9137 ± 0.0012 against 0.9099 ± 0.0060 , block-held-out F1 0.857 against 0.860. The input formulation is what buys the improvement; the extra supervision buys nothing we can measure here.

4.3 Recall side

REJECT50 asks whether the candidates the retrieval pipeline discarded contain recoverable positives. Both models recover 7 of the 9 true positives, so the sentence-level baseline is not blind to them; what separates the two is what each lets in alongside. The baseline re-admits 15 of the 32 correct rejections to find those 7, F1 0.452, and the verifier re-admits 4, F1 0.700. Nine positives is a small numerator — the Clopper–Pearson interval on 7/9 is $[0.40, 0.97]$ — so we read this qualitatively: a curator mining the discarded candidates with the verifier reviews 11 candidates of which 4 are wrong, rather than 22 of which 15 are.

4.4 An operating curve, not a verdict

The verifier’s precision moves from 0.779 to 0.947 as τ moves (Table 4), and the right point on that curve is not a property of the model. Ingestion into a reference database pays for false positives and belongs at the high end; assembling candidates for

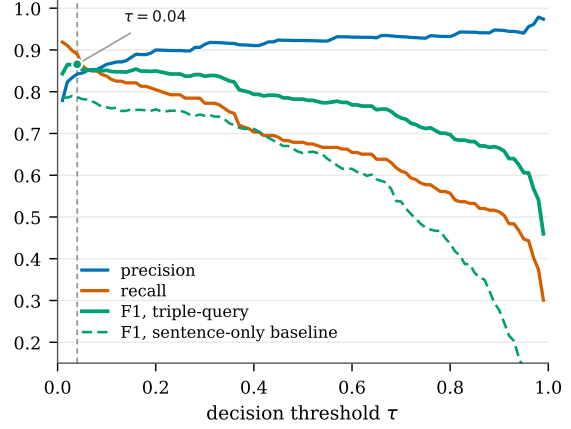


Figure 1: Precision, recall and F1 of the verifier against the decision threshold on all 437 rows, with the sentence-level baseline’s F1 for comparison (same corpus and encoder). The marked threshold is block-held-out. The verifier dominates at every τ .

expert review pays for false negatives and belongs at the low end. A filter that emits a verdict fixes that choice at training time, for users who were never asked. A calibrated score with an adjustable threshold leaves it to them (Geifman and El-Yaniv, 2017).

5 What the remaining errors are made of

The verifier’s 47 false positives at its block-held-out operating point are not homogeneous. Read by hand, they sort into a small number of mechanisms (Table 5): *pair-binding*, where both taxa and the relation are correct but the relation’s argument is a third organism; *co-listing*, where the two taxa are members of one enumeration and interact with neither each other nor anything in particular; *negation*, where the passage asserts the opposite; *same organism*, where the two “taxa” are one organism named twice or an organism and its own clade; candidates whose relation term names no biotic interaction at all (*relative to*, *hybridization*, *migration*); and *entity failures*, where a taxon string is a taxonomic author, an organ, or an adjective inside a pathogen’s name (“*equine influenza virus*”). A further handful read as gold errors and are discussed in the Limitations.

A bound on component-wise verification. The pair-binding class has a structural property worth stating, because it rules out the obvious alternative architecture. Its members have both taxa correctly resolved *and* the correct relation; what is wrong is a binding that exists only in retrieval’s combinatorics. A verifier that checks s_1 , r and s_2 separately

Mechanism	Candidate triple	Passage (abridged)
Pair-binding	(<i>T. gondii</i> , parasite of, swine)	“The entero-pathogenic <i>Cystoisospora suis</i> , a coccidian parasite of swine and close relative to <i>Toxoplasma gondii</i> ...”
Co-listing	(katyids, prey, beetles)	“... remains were from insects ... (e.g. beetles , dragonflies, cockroaches, and female katyids), indicating that eavesdropping on prey signals is not the only prey-finding strategy used by this <i>bat</i> .”
Negation	(wheat, hosts, <i>F. acuminatum</i>)	“... <i>F. acuminatum</i> in New Zealand was found only on non-wheat hosts .”
Same organism	(Chinese knotweed, pollinators, <i>P. chinensis</i>)	“... potential pollinators of Chinese knotweed (<i>Persicaria chinensis</i>).”
Entity	(equine , infections, human)	“... evidence for equine influenza virus infections in human beings.”

Table 5: One example of each mechanism among the verifier’s false positives. The relation’s true argument is italicised where it is not the queried taxon; in the entity case an adjective inside a pathogen’s name was matched as the host.

System	P	R	F1	fixed	broken
Verifier	0.825	0.898	0.860	—	—
+ 7 rules	0.859	0.890	0.874	11	2
+ 8 rules	0.878	0.878	0.878	17	5

Table 6: Deterministic candidate rules applied before the verifier’s decision, all 437 rows, block-held-out thresholds. “Fixed” and “broken” are relative to the verifier alone; McNemar $p = 0.027$ (7 rules) and 0.019 (8). Both rule sets improve every block.

— however good each check is — cannot see it. On the BIOTX100 candidates, where triple-level grades make the classes countable, 16 of the 35 unsupported candidates are pair-binding failures (the rest: 10 entity, 9 predicate). A component-wise verifier accepting a fraction r of the 65 supported triples therefore also accepts all 16, so its precision is at most $65r/(65r + 16)$: 0.802 at full recall. Only a decision that reads the pair *in context* can go above it.

6 Candidate rules

Several of these mechanisms are not questions about the passage at all but about the candidate, and a classifier is a poor instrument for them: whether two strings name one organism, whether a word is a surname, whether “relative to” is an interaction. We answer those with eight deterministic rejection rules applied to the candidate before the verifier decides. *same_organism* (apposition of a common and a scientific name, or identical forms), *own_clade* (an adjacent clade annotation such as “(Acari: Prostigmata)”), and *authority* (a taxonomic author parsed as a taxon) detect the same-organism and entity classes. *non_biotic_relation* rejects phylogenetic, comparative and biogeographic

relation terms; *negated* rejects the two negation forms whose scope provably contains one of the pair (“non-*s* hosts”, “no infections with ... *s*”); *not_an_organism* rejects organs and syndromes; *pathogen_modifier* rejects an adjective that only ever occurs inside a pathogen’s name; and *co_listed* rejects two taxa that are conjuncts of one dependency-parsed list not governed by an interaction noun (“interactions between *A* and *B*” is kept).

How they were validated, and what that licenses. No rule has a parameter fitted to evaluation labels, but rules can still be fitted by *choice*, and four of these — *non_biotic_relation*, *negated*, *not_an_organism* and *pathogen_modifier* — were written after reading the false positives above. To keep their selection from simply memorising the benchmark, every rule had to pass a criterion fixed before it was evaluated: on the 42,747 training rows, at least 85% of the candidates it rejects must be teacher-negative, matching the teacher’s own accuracy. Two first drafts failed it and were narrowed on training data alone — a general negation cue fired on “without affecting the host”, and a role-noun list rejected “viruses” and “fungi”, which are legitimate coarse taxa — and the final rules reject teacher-negatives at 0.88 to 1.00 (*co_listed*, which needs a dependency parse, on a fixed 6,000-row sample). The rules were then frozen, recorded by hash, and evaluated once.

The result is in Table 6: precision rises from 0.825 to 0.878 at a recall cost of 0.020, 17 items fixed and 5 broken, McNemar $p = 0.019$. We report it as an upper estimate, since half the rules were chosen with these errors in view; what the training-data criterion establishes is that they reject

what they are meant to reject on 42k rows they were not chosen on, not that their effect here would replicate exactly on new data. The three rules written before this error analysis, from an earlier one, remove one false positive and one true positive here, which is the honest measure of what rules carry over without being looked for.

7 The deployed configuration

In the pipeline the verifier runs as one 110M-parameter model with two heads: the triple-conditioned verification head described here, and a direction head that decides which taxon is the subject, trained jointly with it and taking the pair in a canonical order so that the direction answer cannot change when the pipeline presents the pair the other way round. Its interaction threshold, 0.5, is fixed on its own development split. On the 437 rows it reaches AUPRC 0.934 and precision 0.852, recall 0.959, F1 0.902; with the candidate rules, precision 0.870, recall 0.951, F1 0.909.¹ It needs no GPU: on eight CPU threads it scores 32.7 candidates per second with both heads, after a one-off 7.5 s to load, which is faster than the pipeline retrieves them, so every candidate is verified rather than a sample.

8 Where the reformulation stops

Entity disambiguation, and a gate that cannot fix it. The natural remedy for entity errors is to gate candidates on taxonomic validity. Against the Open Tree Taxonomy (Rees and Cranston, 2017) such a gate catches 1 of the 35 unsupported BIOTX100 candidates and discards 5 of the 65 supported ones. The reason is diagnostic: *Illiger*, *Dikarya*, *copia* and *sulla* — the surnames, clades, transposon families and vernacular names behind the entity failures — all *resolve* in the taxonomy. The failure is disambiguation, of the kind species-name normalisers address (Gerner et al., 2010), not validity; and the gate’s casualties are viruses, which are legitimate partners in this domain. The *authority* and *same_organism* rules (§6) target the disambiguation directly and are the better instrument.

Argument order is not evaluable on this gold standard. Expert curation marks reversed triples as correct — we verified two by hand, (*Breinlia*,

¹The *co_listed* rule is omitted from this configuration: on the joint model’s accept set it removes one false positive at the cost of three true ones.

host of, *Rattus*) and (*Demidospermus*, parasitized by, *A. inermis*) — so no directional number on this benchmark would measure the model rather than the convention, and the direction head’s evaluation lives on corpora annotated for it. The supervision itself is uneven in a revealing way: among the 5,591 reversed training rows, the rate at which the teacher rejects a reversal is governed by the relation’s *surface string*. Direction-encoding particles and prepositional forms are rejected at 0.51–0.70 (*affecting* 0.70, *affected by* 0.61, *parasite of* 0.57), bare relational nouns at 0.04–0.10 (*host of* 0.08, *pollinators* 0.04). Whether supervision encodes order at all depends on which string retrieval happened to match.

9 Related work

Classifiers that answer an easier question. That a classifier can score well by exploiting how its data were assembled is among the best-established results in NLP. Partial-input baselines recover much of the label from the hypothesis alone in natural language inference (Gururangan et al., 2018; Poliak et al., 2018) and from the passage alone in reading comprehension (Kaushik and Lipton, 2018), and such rules hold in distribution while failing under deployment (Geirhos et al., 2020; Du et al., 2024); the closest task-level precedent is fact verification, where a claim-only classifier matches evidence-aware models because of how the corpus was built (Schuster et al., 2019). Our sentence-level baseline is a partial-input model in exactly this sense, and the specific feature here is that the missing input is the query the pipeline itself issued rather than an annotator artefact (Wiegand et al., 2019; Parmar et al., 2023). Feng et al. (2019) caution that partial-input results license claims about models and corpora rather than about tasks, and we keep ours to the former.

The argument-ignoring failure is already named. Distant supervision assumes that a sentence containing two related entities expresses the relation (Mintz et al., 2009), and its standard relaxation exists because it does not (Riedel et al., 2010). Since TACRED an instance has been the triple (sentence, e_1 , e_2) rather than the sentence (Zhang et al., 2017), and how the pair is injected into the encoder is worth several F1 points (Baldini Soares et al., 2019; Wu and He, 2019; Zhou and Chen, 2022). Rosenman et al. (2020) name our failure exactly, as an “event heuristic” that classifies on (sentence,

relation) while ignoring the arguments. *We claim no part of that diagnosis.* What is ours is a controlled measurement of what it costs in a verification setting built by co-occurrence retrieval, located by block; a bound on component-wise remedies; and the candidate rules. The entity-type shortcut usually invoked for such errors (Peng et al., 2020; Mtumbuka and Schockaert, 2024) is unavailable here, since both slots are filled from one inventory of taxon surface forms. Error attribution across pipeline stages is standard and easy to get wrong (Taillé et al., 2020; Ai and Kavuluru, 2023).

Conditioning on the query. Casting a relation as a question about a named entity (Levy et al., 2017), or as an entailment hypothesis verbalising the relation between two of them (Obamuyide and Vlachos, 2018; Sainz et al., 2021), is the reformulation our input format instantiates, and the resulting decision — does this passage support this specific claim — has the shape of scientific claim verification (Wadden et al., 2020). Distilling a large model’s judgments into a small student is established practice (West et al., 2022; Gekhman et al., 2023). Selective classification thresholds a score instead of issuing a verdict (Geifman and El-Yaniv, 2017), which is how extraction is made useful to human abstractors: Swaminathan et al. (2024) abstain under calibrated thresholds to automate 57–95% of clinical chart abstraction at controlled error.

Text mining for interaction databases. GloBI (Poelen et al., 2014) is populated in part by text mining (Dimitrova et al., 2020). Judging an extracted fact before it reaches a curator is an old idea (Rodriguez-Esteban et al., 2006), as is evaluating such tools by whether they help the curator (Hirschman et al., 2012; Britan et al., 2018; Lee et al., 2018). Cuzick et al. (2023), independently, take the interacting pair rather than the passage as the annotation unit — the unit our pair-binding class isolates. Large language models are applied to interaction extraction at scale (Keck et al., 2025; Farrell et al., 2024; D’Souza et al., 2025; Zou et al., 2026), but as extractors: Zou et al. (2026) standardise names and categories post hoc without checking extracted interactions against the text that produced them. We are not aware of published work that verifies literature-mined biotic-interaction triples against their source passage before ingestion.

10 Conclusion

When candidates are retrieved by co-occurrence, the sentence-level question is answered before it is asked, and a classifier built for it is left with nothing to decide. Asking instead whether the passage supports the candidate — with the same encoder and the same data — raises AUPRC from 0.851 to 0.910, and the gain sits exactly where passages name more than one pair. The errors that remain are mostly questions about the candidate rather than the passage, and deterministic rules answer those better than a classifier does. The result is a calibrated score where a verdict stood, cheap enough to run on every candidate.

Limitations

One benchmark, singly annotated. 437 rows in three blocks, every label assigned by one domain expert, no item graded twice: we report no inter-annotator agreement and cannot bound label noise. Several of the verifier’s most confident false positives read as gold errors — a bat tick “reported . . . to occasionally feed on humans”, labelled negative. We have changed none of them: correcting gold after seeing model output biases the evaluation toward the model, so they go instead to blind expert review, mixed with an equal number of random controls and with model scores hidden. On the recall side the denominator is 9 positives.

The rules are partly fitted by choice. Four of the eight rules were selected after reading this benchmark’s false positives (§6). The training-data criterion bounds how much that selection can have memorised the benchmark, but it does not remove it, and the rules’ $p = 0.019$ should be read as optimistic until it is confirmed on candidates neither the rules nor their author have seen. The rules also fire on no row of a separate 100-candidate development sample, so that sample can neither confirm nor contradict them.

Fresh candidates are harder. On that development sample — 100 retrieved candidates drawn after the benchmark, half of them from passages containing several candidate pairs — the verifier reaches F1 0.698, well below its benchmark figure. Its label provenance is not identical to the benchmark’s, so we do not read this as a measured drop, but it is the clearest sign that multi-pair passages remain the hard case and that the benchmark may understate their share of a live pool.

The diagnosis is triple-level; the evaluation is pair-level. Only BIOTX100 carries whole-triple gold. Evaluating at the pair level everywhere means a candidate with the right pair and a misbound relation counts as correct, so the headline understates what triple conditioning buys; triple-level gold on the other two blocks would sharpen it.

Teacher noise and pretraining exposure. The training labels come from a prompted teacher wrong about one time in seven, and the student inherits that boundary; training on model-generated labels can degrade encoders (Lu and Smith, 2025). The passages are published MEDLINE text and the teacher is a pretrained model, so we cannot exclude that it saw them, and its labels may be optimistic for genuinely new literature.

Upstream failures are out of reach. Of the 50 candidates the retrieval pipeline discarded, 32 carry an incorrect interaction term and 17 a mis-resolved taxon, 14 of them both. Those fail before any verifier is consulted, and no change to the decision stage, ours included, addresses them.

One system, one domain. One retrieval system, one interaction vocabulary, taxon–taxon interactions in MEDLINE. The mechanism — a pair-agnostic decision over passages that name several candidate pairs — is not specific to ours, but we have not shown that other pools fail the same way.

References

- Xuguang Ai and Ramakanth Kavuluru. 2023. End-to-end models for chemical–protein interaction extraction: Better tokenization and span-based pipeline strategies. In *2023 IEEE 11th International Conference on Healthcare Informatics (ICHI)*, pages 610–618. IEEE. ArXiv:2304.01344.
- Livio Baldini Soares, Nicholas FitzGerald, Jeffrey Ling, and Tom Kwiatkowski. 2019. [Matching the blanks: Distributional similarity for relation learning](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2895–2905, Florence, Italy. Association for Computational Linguistics.
- Aurore Britan, Isabelle Cusin, Valérie Hinard, Luc Motin, Emilie Pasche, Julien Gobeill, Valentine Rech de Laval, Anne Gleizes, Daniel Teixeira, Pierre-André Michel, Patrick Ruch, and Pascale Gaudet. 2018. [Accelerating annotation of articles via automated approaches: evaluation of the neXtA5 curation-support tool by neXtProt](#). *Database*, 2018: bay129.

- Alayne Cuzick, James Seager, Valerie Wood, Martin Urban, Kim Rutherford, and Kim E. Hammond-Kosack. 2023. [A framework for community curation of inter-species interactions literature](#). *eLife*, 12:e84658.
- Mariya Dimitrova, Jorrit Poelen, Georgi Zhelezov, Teodor Georgiev, Donat Agosti, and Lyubomir Penev. 2020. [Semantic publishing enables text mining of biotic interactions](#). *Biodiversity Information Science and Standards*, 4:e59036.
- Jennifer D’Souza, Zachary Laubach, Tarek Al Mustafa, Sina Zarrieß, Robert Frühstückl, and Phyllis Illari. 2025. [Mining for species, locations, habitats, and ecosystems from scientific papers in invasion biology: A large-scale exploratory study with large language models](#). In *Proceedings of the 1st Workshop on Ecology, Environment, and Natural Language Processing (NLP4Ecology2025)*, pages 16–23, Tallinn, Estonia. University of Tartu Library.
- Mengnan Du, Fengxiang He, Na Zou, Dacheng Tao, and Xia Hu. 2024. [Shortcut learning of large language models in natural language understanding](#). *Communications of the ACM*, 67(1):110–120.
- Maxwell J. Farrell, Nicolas Le Guillarme, Liam Brierley, Bronwen Hunter, Daan Scheepens, Anna Willoughby, Andrew Yates, and Nicole Mideo. 2024. [The changing landscape of text mining: a review of approaches for ecology and evolution](#). *Proceedings of the Royal Society B: Biological Sciences*, 291(2027):20240423.
- Shi Feng, Eric Wallace, and Jordan Boyd-Graber. 2019. [Misleading failures of partial-input baselines](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5533–5538, Florence, Italy. Association for Computational Linguistics.
- Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*, pages 4878–4887.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. [Shortcut learning in deep neural networks](#). *Nature Machine Intelligence*, 2(11):665–673.
- Zorik Gekhman, Jonathan Herzig, Roei Aharoni, Chen Elkind, and Idan Szpektor. 2023. [TrueTeacher: Learning factual consistency evaluation with large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2053–2070, Singapore. Association for Computational Linguistics.
- Martin Gerner, Goran Nenadic, and Casey M. Bergman. 2010. [LINNAEUS: A species name identification system for biomedical literature](#). *BMC Bioinformatics*, 11:85.
- Julien Gobeill, Déborah Caucheteur, Pierre-André Michel, Luc Mottin, Emilie Pasche, and Patrick Ruch.

672	2020. SIB literature services: RESTful customizable search engines in biomedical literature, enriched with automatically mapped biomedical concepts . <i>Nucleic Acids Research</i> , 48(W1):W12–W16.	729
673		730
674		
675		
676	Suchin Gururangan, Swabha Swayamdipta, Omer Levy,	
677	Roy Schwartz, Samuel R. Bowman, and Noah A.	
678	Smith. 2018. Annotation artifacts in natural lan-	
679	guage inference data . In <i>Proceedings of the 2018</i>	
680	<i>Conference of the North American Chapter of the</i>	
681	<i>Association for Computational Linguistics: Human</i>	
682	<i>Language Technologies, Volume 2 (Short Papers)</i> ,	
683	pages 107–112, New Orleans, Louisiana. Associa-	
684	tion for Computational Linguistics.	
685	Lynette Hirschman, Gully A. P. C. Burns, Martin	
686	Krallinger, Cecilia Arighi, K. Bretonnel Cohen,	
687	Alfonso Valencia, Cathy H. Wu, Andrew Chatr-	
688	Aryamontri, Karen G. Dowell, Eva Huala, Anália	
689	Lourenço, Robert Nash, Anne-Lise Veuthey, Thomas	
690	Wiegiers, and Andrew G. Winter. 2012. Text	
691	mining for the biocuration workflow . <i>Database</i> ,	
692	2012:bas020.	
693	Divyansh Kaushik and Zachary C. Lipton. 2018. How	
694	much reading does reading comprehension require?	
695	A critical investigation of popular benchmarks . In	
696	<i>Proceedings of the 2018 Conference on Empirical</i>	
697	<i>Methods in Natural Language Processing</i> , pages	
698	5010–5015, Brussels, Belgium. Association for Com-	
699	putational Linguistics.	
700	François Keck, Henry Broadbent, and Florian Altermatt.	
701	2025. Extracting massive ecological data on state and	
702	interactions of species using large language models .	
703	<i>bioRxiv</i> . Preprint, posted 27 January 2025.	
704	Kyubum Lee, Maria Livia Famiglietti, Aoife McMa-	
705	hon, Chih-Hsuan Wei, Jacqueline Ann Langdon	
706	MacArthur, Sylvain Poux, Lionel Breuza, Alan	
707	Bridge, Fiona Cunningham, Ioannis Xenarios, and	
708	Zhiyong Lu. 2018. Scaling up data curation using	
709	deep learning: An application to literature triage in	
710	genomic variation resources . <i>PLOS Computational</i>	
711	<i>Biology</i> , 14(8):e1006390.	
712	Omer Levy, Minjoon Seo, Eunsol Choi, and Luke	
713	Zettlemoyer. 2017. Zero-shot relation extraction via	
714	reading comprehension . In <i>Proceedings of the 21st</i>	
715	<i>Conference on Computational Natural Language</i>	
716	<i>Learning (CoNLL 2017)</i> , pages 333–342, Vancouver,	
717	Canada. Association for Computational Linguistics.	
718	Yucheng Lu and Kazimier Smith. 2025. Feeding LLM	
719	annotations to BERT classifiers at your own risk.	
720	<i>arXiv preprint arXiv:2504.15432</i> .	
721	Mike Mintz, Steven Bills, Rion Snow, and Dan Juraf-	
722	sky. 2009. Distant supervision for relation extrac-	
723	tion without labeled data. In <i>Proceedings of ACL-</i>	
724	<i>IJCNLP</i> , pages 1003–1011.	
725	Frank Martin Mtumbuka and Steven Schockaert. 2024.	
726	Entity or relation embeddings? an analysis of encod-	
727	ing strategies for relation extraction . In <i>Findings</i>	
728	<i>of the Association for Computational Linguistics:</i>	
	<i>EMNLP 2024</i> , pages 6003–6022, Miami, Florida,	729
	USA. Association for Computational Linguistics.	730
	Rodrigo Nogueira and Kyunghyun Cho. 2019. Pas-	731
	sage re-ranking with BERT . <i>arXiv preprint</i>	732
	<i>arXiv:1901.04085</i> .	733
	Abiola Obamuyide and Andreas Vlachos. 2018. Zero-	734
	shot relation classification as textual entailment . In	735
	<i>Proceedings of the First Workshop on Fact Extraction</i>	736
	<i>and VERification (FEVER)</i> , pages 72–78, Brussels,	737
	Belgium. Association for Computational Linguistics.	738
	Mihir Parmar, Swaroop Mishra, Mor Geva, and Chitta	739
	Baral. 2023. Don’t blame the annotator: Bias al-	740
	ready starts in the annotation instructions . In <i>Pro-</i>	741
	<i>ceedings of the 17th Conference of the European</i>	742
	<i>Chapter of the Association for Computational Lin-</i>	743
	<i>guistics</i> , pages 1779–1789, Dubrovnik, Croatia. As-	744
	sociation for Computational Linguistics.	745
	Hao Peng, Tianyu Gao, Xu Han, Yankai Lin, Peng	746
	Li, Zhiyuan Liu, Maosong Sun, and Jie Zhou. 2020.	747
	Learning from context or names? an empirical study	748
	on neural relation extraction . In <i>Proceedings of the</i>	749
	<i>2020 Conference on Empirical Methods in Natural</i>	750
	<i>Language Processing (EMNLP)</i> , pages 3661–3672,	751
	Online. Association for Computational Linguistics.	752
	Jorrit H Poelen, James D Simons, and Chris J Mungall.	753
	2014. Global biotic interactions: An open infrastruc-	754
	ture to share and analyze species-interaction datasets.	755
	<i>Ecological Informatics</i> , 24:148–159.	756
	Adam Poliak, Jason Naradowsky, Aparajita Haldar,	757
	Rachel Rudinger, and Benjamin Van Durme. 2018.	758
	Hypothesis only baselines in natural language infer-	759
	ence . In <i>Proceedings of the Seventh Joint Confer-</i>	760
	<i>ence on Lexical and Computational Semantics</i> , pages	761
	180–191, New Orleans, Louisiana. Association for	762
	Computational Linguistics.	763
	Qwen Team. 2025. Qwen3 technical report. <i>arXiv</i>	764
	<i>preprint arXiv:2505.09388</i> .	765
	Jonathan A. Rees and Karen Cranston. 2017. Auto-	766
	mated assembly of a reference taxonomy for phylo-	767
	genetic data synthesis . <i>Biodiversity Data Journal</i> ,	768
	5:e12581.	769
	Sebastian Riedel, Limin Yao, and Andrew McCallum.	770
	2010. Modeling relations and their mentions with-	771
	out labeled text . In <i>Machine Learning and Knowl-</i>	772
	<i>edge Discovery in Databases (ECML PKDD 2010)</i> ,	773
	Lecture Notes in Computer Science, pages 148–163,	774
	Barcelona, Spain. Springer Berlin Heidelberg.	775
	Raul Rodriguez-Esteban, Ivan Iossifov, and Andrey	776
	Rzhetsky. 2006. Imitating manual curation of text-	777
	mined facts in biomedicine . <i>PLoS Computational</i>	778
	<i>Biology</i> , 2(9):e118.	779
	Shachar Rosenman, Alon Jacovi, and Yoav Goldberg.	780
	2020. Exposing shallow heuristics of relation ex-	781
	traction models with challenge data . In <i>Proceed-</i>	782
	<i>ings of the 2020 Conference on Empirical Methods</i>	783

in *Natural Language Processing (EMNLP)*, pages 3702–3710, Online. Association for Computational Linguistics.

Patrick Ruch, Emilie Pasche, Pierre-André Michel, and Déborah Caucheteur. 2024. [BiotXplorer: Navigating evidence-based biotic interactions](#). *Biodiversity Information Science and Standards*, 8:e135453.

Oscar Sainz, Oier Lopez de Lacalle, Gorka Labaka, Ander Barrena, and Eneko Agirre. 2021. [Label verbalization and entailment for effective zero and few-shot relation extraction](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 1199–1212, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Tal Schuster, Darsh Shah, Yun Jie Serene Yeo, Daniel Roberto Filizzola Ortiz, Enrico Santus, and Regina Barzilay. 2019. [Towards debiasing fact verification models](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3419–3425, Hong Kong, China. Association for Computational Linguistics.

Akshay Swaminathan, Ivan Lopez, William Wang, Ujwal Srivastava, Edward Tran, Aarohi Bhargava-Shah, Janet Y. Wu, Alexander L. Ren, Kaitlin Caoili, Brandon Bui, Layth Alkhani, Susan Lee, Nathan Mohit, Noel Seo, Nicholas Macedo, Winson Cheng, Charles Liu, Reena Thomas, Jonathan H. Chen, and Olivier Gevaert. 2024. [Selective prediction for extracting unstructured clinical data](#). *Journal of the American Medical Informatics Association*, 31(1):188–197.

Bruno Taillé, Vincent Guigue, Geoffrey Scoutheeten, and Patrick Gallinari. 2020. [Let’s stop incorrect comparisons in end-to-end relation extraction!](#) In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3689–3701. Association for Computational Linguistics.

David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. 2020. [Fact or fiction: Verifying scientific claims](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7534–7550, Online. Association for Computational Linguistics.

Peter West, Chandra Bhagavatula, Jack Hessel, Jena D. Hwang, Liwei Jiang, Ronan Le Bras, Ximing Lu, Sean Welleck, and Yejin Choi. 2022. [Symbolic knowledge distillation: from general language models to commonsense models](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4602–4625, Seattle, United States. Association for Computational Linguistics.

Michael Wiegand, Josef Ruppenhofer, and Thomas Kleinbauer. 2019. [Detection of abusive language: the problem of biased datasets](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 602–608, Minneapolis, Minnesota. Association for Computational Linguistics.

Shanchan Wu and Yifan He. 2019. [Enriching pre-trained language model with entity information for relation classification](#). In *Proceedings of the 28th ACM International Conference on Information and Knowledge Management (CIKM)*, pages 2361–2364, Beijing, China. Association for Computing Machinery.

Yuhao Zhang, Victor Zhong, Danqi Chen, Gabor Angeli, and Christopher D. Manning. 2017. [Position-aware attention and supervised data improve slot filling](#). In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, pages 35–45, Copenhagen, Denmark. Association for Computational Linguistics.

Wenxuan Zhou and Muhao Chen. 2022. [An improved baseline for sentence-level relation extraction](#). In *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 161–168, Online only. Association for Computational Linguistics.

Heng-Xing Zou, Xiaohao Yang, Thabassum H. Hajamaideen, Olivia J. Stein, Roxanne S. Beltran, Benjamin G. Freeman, Mark Lindquist, Eliot T. Miller, Summer Mengarelli, Charlotte M. Probst, Fernanda S. Valdovinos, Derek Van Berkel, Phoebe L. Zarnetske, Brian C. Weeks, and Kai Zhu. 2026. [Large language models unlock the ecology of species interactions](#). *bioRxiv*. Preprint, posted 10 February 2026.

A Per-seed results

	triple-conditioned		sentence-level	
	AUPRC	F1 @ 0.5	AUPRC	
seed 1	0.9140	0.7832	0.8578	
seed 2	0.9030	0.8018	0.8513	
seed 3	0.9128	0.7755	0.8449	
mean ± sd	0.9099 ± 0.0060	0.7868 ± 0.0136	0.8513 ± 0.0064	0.8513
ensemble	0.9123	0.7822	—	—

Table 7: Per-seed figures. The ensemble averages three checkpoints’ probabilities and is what the body of the paper reports at operating points.

B Negative results and cost

Table 8 collects interventions that did not pay, because each is a plausible first thing to try. Self-distillation at equal capacity is null, suggesting

Intervention	Benefit	Cost
Taxon-validity gate	1 of 35 unsupported caught	−5/65 supported; rejects viruses
Self-distillation	none	−0.001 P, −0.008 F1
6-layer compression	1.9× CPU throughput	−0.088 precision
Corpus signal-matching	none	−0.059 AUPRC
Rule-family training data	none	AUPRC +0.005 to −0.037

Table 8: Five plausible design choices that measurably do not pay.

the teacher’s knowledge is already saturated in the student. Compression to six layers buys throughput the pipeline does not need for 0.088 precision. Reweighting the corpus toward the deployment signal distribution cost 0.059 AUPRC. Most instructive is the last row: before the candidate rules existed, we turned the same mechanisms into *training data* — synthetic negatives for the same-organism, own-clade and authority classes, and in a second variant for co-listing — and retrained. The first variant left AUPRC unchanged within seed noise and the second cost 0.037. Mechanisms that are properties of the candidate are better answered by a rule than taught to a classifier. Deltas in the first four rows were measured under an earlier version of the evaluation and are given as directions.